

Chapter 13 Homework Answer Sheet

Name: _____ Student ID: _____ Instructor: _____ Score: _____

Part A multiple choice (15 points)

1-5: CDBAD 6-10: DBDDC 11-15: DDCBD

Part B short question (20 points)

16. (4 points)

(a) $\text{Ar} < \text{Kr} < \text{Xe}$

(b) As molar mass and **atomic size** of the ideal gas **increases**, the **electron cloud** is more **easily polarizable**. The **greater** the **polarizability**, the **stronger** the **dispersion** forces between the gas atoms and water, leading to **greater solubility** of the atom.

17. (3 points)

(a) Endothermic.

(b) The increase of entropy favors the formation of the solution.

18. (6 points)

(a)

$$X_{\text{CH}_3\text{OH}} = \frac{\text{mole of CH}_3\text{OH}}{\text{mole of CH}_3\text{OH} + \text{mole of H}_2\text{O}} = \frac{14.6 \div 32.04}{184 \div 18.015 + 14.6 \div 32.04} = 0.0427$$

(b)

$$\% \text{CH}_3\text{OH by mass} = \frac{14.6}{14.6 + 184} \times 100\% = 7.35\%$$

(c)

$$\text{molality of CH}_3\text{OH} = \frac{\text{mole of CH}_3\text{OH}}{\text{H}_2\text{O kg}} = \frac{14.6 \div 32.04 \text{ mol}}{0.184 \text{ kg}} = 2.48 \text{ m}$$

19. (4 points)

(a)

$$\text{mole of C}_3\text{H}_8\text{O}_3 = 28.5 \text{ g} \times \frac{1 \text{ mol}}{92.094 \text{ g}} = 0.3095 = 0.310 \text{ mol}$$

$$\text{mole of H}_2\text{O} = 125 \text{ g} \times \frac{1 \text{ mol}}{18.015 \text{ g}} = 6.939 = 6.94 \text{ mol}$$

$$x_{\text{H}_2\text{O}} = \frac{6.939}{0.3095 + 6.939} = 0.9573 = 0.957$$

$$P_{\text{H}_2\text{O}} = x_{\text{H}_2\text{O}} P_{\text{H}_2\text{O}}^0 = 0.9573 \times 233.7 = 223.7 = 224 \text{ torr}$$

(b)

$$\Delta P_{\text{C}_2\text{H}_5\text{OH}} = x_{\text{C}_2\text{H}_6\text{O}_2} P_{\text{C}_2\text{H}_5\text{OH}} = x_{\text{C}_2\text{H}_6\text{O}_2} \times 1.00 \times 10^2 \text{ torr} = 10.0 \text{ torr}$$

$$x_{\text{C}_2\text{H}_6\text{O}_2} = \frac{10.0 \text{ torr}}{1.00 \times 10^2 \text{ torr}} = 0.100$$

$$\text{moles of C}_2\text{H}_5\text{OH} = 1.00 \times 10^3 \text{ g} \times \frac{1 \text{ mol}}{46.069 \text{ g}} = 21.71 = 21.7 \text{ mol}$$

$$x_{\text{C}_2\text{H}_6\text{O}_2} = \frac{\text{mole of C}_2\text{H}_6\text{O}_2}{\text{mole of C}_2\text{H}_5\text{OH} + \text{mole of C}_2\text{H}_6\text{O}_2} = \frac{\text{mole of C}_2\text{H}_6\text{O}_2}{\text{mole of C}_2\text{H}_5\text{OH} + 21.71 \text{ mol}} = 0.100$$

$$\text{moles of C}_2\text{H}_6\text{O}_2 = 2.412 = 2.41 \text{ mol}$$

$$\text{mass of C}_2\text{H}_6\text{O}_2 = 2.412 \text{ mol} \times \frac{62.068 \text{ g}}{1 \text{ mol}} = 149.7 = 150. \text{ g}$$

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20. (3 points)

(a) A 0.10 *m* aqueous solution of NaCl have a **higher** boiling point.

Boiling-point elevation is directly related to total moles of dissolved particles. NaCl is a strong electrolyte, The 0.10 *m* NaCl produces twice as many dissolved particles as 0.10 *m* C₆H₁₂O₆. So its boiling point is higher than that of 0.10 *m* C₆H₁₂O₆.

(b)

In solutions of strong electrolytes like NaCl, **ion pairing** reduces the effective number of particles in solution, decreasing the change in boiling point. The actual boiling point is then lower than the calculated boiling point for a 0.10 *m* solution.